

AGRILINK: SMART FARM-TO-CUSTOMER SUPPLY CHAIN

Comprehensive Project Handover, Technical Architecture & Junior Student Onboarding Guide

Faculty Project Invitation & Technical Roadmap | Academic Year 2026

■ PROJECT HANDOVER BRIEF

Invitation to Junior Students: AgriLink is an active, fully integrated multi-tier agriculture platform bridging smallholder farmers directly to institutional and retail buyers. This document details everything built so far, the system architecture, and specific high-impact modules open for junior engineers, researchers, and makerspace developers to take forward for Capstones, Hackathons, and Publications.

1 Project Title

Full Title: AgriLink: An Intelligent Machine Learning Driven Farm-to-Customer Supply Chain & Dynamic Listing Recommendation Platform

Short Name / Acronym: AgriLink (v1.0.0)

Keywords: Direct Farm-to-Fork, Hybrid LightFM Recommendation, Perishable Supply Chain, Flutter Web/Mobile, PostgreSQL JSON Views, FastAPI Microservices, IoT Freshness Tracking.

2 Problem Statement

Agricultural distribution in developing economies suffers from acute systemic bottlenecks:

- **Multi-Tier Intermediary Exploitation:** 4 to 6 tiers of middlemen inflate consumer costs by up to 60% while farmers receive only 20–30% of final retail value.
- **Post-Harvest Losses & Perishability:** Inefficient scheduling and lack of direct demand linkage result in 25–35% of fresh fruits and vegetables perishing before market arrival.
- **Information Asymmetry:** Farmers lack real-time market demand and price transparency; institutional buyers (hostels, hospitals, restaurants) struggle with erratic supply and opaque quality grading.
- **Listing Incoherence in Generic E-Commerce:** Existing platforms recommend generic crop types (e.g., 'Tomato') rather than dynamic, actionable seller batches characterized by freshness age, harvest distance, and farmer reputation.

3 Motivation / Why This Project?

Agriculture is the economic foundation for over 50% of the national workforce. Inspiring junior engineers to work on AgriLink offers rare multi-dimensional benefits:

- **Direct Socio-Economic Impact:** Disintermediation puts profits directly back into rural farming families and cooperative aggregator hubs.
- **Cutting-Edge Engineering Exposure:** Practical intersection of Hybrid Collaborative Filtering (LightFM), high-throughput asynchronous APIs (FastAPI), modern UI design (Flutter), and native database JSON performance.
- **Competition & Hackathon Ready:** Aligns perfectly with national hackathons (Smart India Hackathon, AgTech Sprints) and peer-reviewed research publications in Agro-Informatics and Applied AI.

4 Proposed Solution

AgriLink delivers an end-to-end, multi-stakeholder agricultural ecosystem linking Farmers, Rural Aggregators, Commercial/Retail Buyers, and Delivery Logistics Partners:

1. **Specific Seller-Listing Recommendations:** Formulates the recommendation problem as predicting buyer affinity for specific active listings parameterized by harvest freshness percentage, spatial delivery distance, and organic certification.
2. **Multi-Role Digital Experience:** Dedicated interfaces for (a) Farmers (inventory management, mandi pricing, harvest logs), (b) Aggregator Hubs (quality inspection, batch consolidation), (c) Buyers (smart cart, group buying, recurring meal schedules), and (d)

Delivery Riders (turn-by-turn route dispatch).

3. **High-Performance Native JSON Pipeline:** Replaces heavy ORM latency with PostgreSQL server-side JSON aggregation views, delivering response times under 45ms.

5 Technology & Components Involved

Component Layer	Technologies / Tools / Frameworks	Purpose / Application
Frontend Client	Flutter 3.44 (Dart), Provider / BLoC, Google Fonts, Material 3	Responsive Web, Android & iOS cross-platform portals for all 4 roles.
AI / ML Engine	Python 3.10+, LightFM, Scikit-Learn, SVD, Pandas, NumPy, Matplotlib	Hybrid recommendation model, WARP pairwise ranking, benchmark suite.
API Gateway & Backend	FastAPI (Python, Uvicorn, Pydantic), Java 21 Spring Boot Microservices	High-throughput RESTful endpoints, CORS middleware, ML inference routing.
Database & Storage	PostgreSQL 17 / Supabase Cloud, JSON Aggregation Views, H2 Dev DB	100+ seeded records, relational constraints, geospatial proximity queries.
Security & Auth	Dynamic 6-Digit OTP Verification, Google OAuth, Role-Based Access Control	Phone-based rural farmer verification and enterprise institutional access.
Makerspace / IoT	ESP32, DHT22, NPK Soil Sensor, Load Cells, Raspberry Pi Camera, GPS	Hardware node for crate freshness tracking, automated weighing & grading.

6 Architecture / System Diagram

The system adopts a decoupled 4-tier microservices architecture designed for extreme responsiveness and edge extensibility:

1. CLIENT LAYER (Flutter Multi-Platform UI)
- Farmer Portal (Listing & Pricing) • Aggregator Hub (QC & Batching) - Customer Portal (Cart, Orders, Smart Feed) • Delivery Agent (Routing & Dispatch)
2. API GATEWAY & MICROSERVICES LAYER
FastAPI Service (Port 8000): Real-time ML Inference, Recommendation Pipelines, Listing APIs Spring Boot Gateway (Port 8080): Enterprise Business Logic, Accounting & Invoice Automation
3. AI / ML RECOMMENDATION ENGINE (LightFM & Hybrid Embeddings)
Utility Score: $S(u, l) = f(p_u, q_l) + \alpha \cdot \text{CosineSim}(c_u, c_l) - \beta \cdot \text{Distance}(u, l) + \gamma \cdot \text{Freshness}(l)$ - Weighted Approximate-Rank Pairwise (WARP) Loss optimizes top-ranked items.
4. PERSISTENCE & IoT DATA LAYER (PostgreSQL 17 / Supabase)
- Native JSON Views: product_listing_view, farmer_dashboard_view, customer_orders_view - IoT Telemetry Ingestion (Temperature, Humidity, Ethylene, Batch GPS Coordinates)

7 Current Status

The core foundations of the AgriLink platform are 100% operational, fully verified, and benchmarked:

- **Backend Infrastructure:** FastAPI REST API (v1.0.0) running on http://127.0.0.1:8000 with CORS support, live health check endpoints, and Supabase cloud connectivity.
- **Database Schema & Seeding:** 22 modular SQL migrations executed; populated with 105+ user entities, 100+ live crop listings across Kolar, Malur, Chickballapur, and Bangalore regions.
- **ML Recommendation Pipeline:** LightFM Hybrid algorithm trained and benchmarked across 120 customer profiles and 50 seller listings with precision metrics recorded in Excel and graphical plots.
- **Frontend Interface:** Multi-screen Flutter UI implementing live listing feeds, smart cart management, dynamic OTP generation, and profile management.

8 Current Prototype & Working Demo

The system can be tested immediately using our live test suite and interactive API endpoints:

Endpoint / Feature	Method & URL / Action	Demonstrated Output / Result
Health Verification	GET /api/v1/health	Returns service healthy, database connected, ml_loaded: true.
Personalized ML Feed	GET /api/v1/recommendations/{buyer_id}	Ranks top 10 specific listings sorted by purchase likelihood & freshness.
Marketplace Listings	GET /api/v1/listings	Retrieves active stock (Tomatoes, Onions, Sharbati Wheat, Mangoes).
Automated Test Suite	python test_live_system.py	Executes end-to-end multi-service test showing sub-35ms response times.

9 Work Completed (Technical Deliverables)

- Database JSON Engine:** Built database views (product_listing_view, farmer_dashboard_view) reducing backend round-trips by 75% and query latencies to <45ms.
- ML Multi-Model Benchmark:** Implemented 4 competing recommendation models. The LightFM WARP model significantly outperformed traditional SVD and Content-Based baselines:

Model Architecture	Precision@10	Recall@10	MAP@10	NDCG@10	AUC-ROC
LightFM (WARP Hybrid)	0.842	0.785	0.812	0.856	0.924
SVD Matrix Factorization	0.724	0.651	0.695	0.738	0.841
Content-Based Cosine	0.615	0.582	0.590	0.624	0.765
Random Baseline	0.185	0.162	0.145	0.198	0.502

- Multi-Role UI/UX:** Developed Flutter screens for dynamic role selection, simulated 6-digit OTP verification, real-time cart computation, and responsive mobile/web layout.
- Academic Dissertation Report:** Generated comprehensive Word & PDF technical documentation detailing database schemas, mathematical formulations, and benchmarking plots.

10 Pending & Future Work (Where Juniors Can Contribute)

This is the roadmap of high-impact feature tracks open for junior teams to lead:

- Track A: Computer Vision & Edge AI Quality Grading:** Build a mobile CV module (YOLOv8-Nano / MobileNetV4) using smartphone camera images to automatically grade produce into A+, A, B, or C quality based on surface blemishes, color uniformity, and size estimation.
- Track B: IoT Cold Chain & Smart Crate Telemetry:** Deploy ESP32 nodes with DHT22 temperature/humidity and ethylene gas sensors inside transport crates to transmit real-time perishability alerts and GPS tracking over MQTT/LoRaWAN.
- Track C: Dynamic Multi-Stop Delivery Route Optimization:** Implement a Vehicle Routing Problem (VRP) algorithm with time windows (using Google OR-Tools or Genetic Algorithms) to optimize aggregator-to-customer delivery routes, minimizing fuel consumption and carbon footprint.
- Track D: Mandi Price Forecasting & Dynamic Pricing:** Develop an LSTM/XGBoost time-series forecasting model using historical Agmarknet mandi prices to recommend optimal selling prices to farmers based on localized supply surges and rainfall patterns.
- Track E: Multilingual Voice Interface for Rural Farmers:** Integrate OpenAI Whisper / AI4Bharat Bhashini voice-to-text API allowing vernacular speech-to-listing (Hindi, Kannada, Telugu, Tamil) so illiterate farmers can list crops effortlessly.
- Track F: Blockchain Batch Provenance & QR Traceability:** Generate verifiable cryptographic QR codes for every crop harvest batch to give consumers transparent 'Farm-to-Fork' origin and pesticide audit trails.

11 Hackathons, Competitions & Research Opportunities

AgriLink is primed for prestigious student competitions and publication venues:

- Smart India Hackathon (SIH):** Direct fit for Ministry of Agriculture & Farmers Welfare problem statements on 'Eliminating Middlemen in Agri-Supply' and 'Smart Logistics'.
- AgTech & SustainTech Competitions:** Ready prototype for AgriThon, NASA Space Apps (Food Security), UN SDG Innovation Challenges, and IEEE Student Project contests.

- **Research Publications:** The empirical comparison of LightFM listing-level ranking vs. classical collaborative filtering is drafted and structured for submission to conferences such as IEEE Access, Springer CCIS, or Agro-Informatics journals.

12 Academic Course Linkage & Project Credits

Students can seamlessly align AgriLink modules with departmental curriculum requirements:

Academic Course / Subject	Suggested Module / Mini-Project / Capstone Deliverable
Machine Learning & Data Science	Tuning LightFM hyperparameters, developing LSTM price forecasting, or building hybrid recommender loss functions.
Computer Vision & Image Processing	Building deep learning segmentation models for fruit/vegetable rot detection and automated visual grading.
Database Management Systems (DBMS)	Designing PostgreSQL spatial indexes (PostGIS), optimizing JSON aggregation views, and sharding order transactions.
Web & Mobile App Development	Enhancing Flutter cross-platform UI, implementing offline SQLite caching, state management, and animations.
Internet of Things (IoT) & Embedded Systems	Prototyping sensor nodes (ESP32/Raspberry Pi) for transit cold-chain monitoring and MQTT telemetry pipelines.
Supply Chain Analytics & Optimization	Modeling VRP logistics algorithms, inventory decay curves, and aggregator hub consolidation networks.

13 Possible Student Roles & Skill Sets Needed

Student Role	Target Skill Sets	Primary Responsibilities in Project
AI / ML Engineer	Python, PyTorch / TF, LightFM, Scikit-Learn, OpenCV	Enhance recommendation algorithms, CV grading models, and price predictor.
Frontend App Developer	Flutter, Dart, Provider / Riverpod, Material UI	Build new screens for delivery tracking, vernacular UI, and offline farmer mode.
Backend / Cloud Engineer	FastAPI, Spring Boot, PostgreSQL, Docker, Redis	Scale microservices, design secure REST/GraphQL endpoints, optimize SQL views.
IoT & Embedded Specialist	ESP32, Arduino, C/C++, MQTT, Sensor Interfacing	Assemble smart transport crate hardware, battery optimization, sensor calibration.
Research & Documentation Lead	Technical Writing, LaTeX, Data Visualization, Benchmarking	Author research papers, maintain repository wikis, lead competition presentations.

14 Current Team & Faculty Mentors

Designation / Role	Name / Department	Contact & Collaboration Channel
Faculty Mentor & Guide	Department Faculty Guide / Project Coordinator	Department Cabin & Official Institutional Email
Makerspace Lab Coordinator	Makerspace & Innovation Lab Mentor	Innovation Lab Office / Hardware Desk
Senior Project Leads	Senior Engineering Team / AgriLink Core Developers	Project GitHub Discussion & Campus Contact
Open Junior Positions	2nd / 3rd Year B.Tech Students (CS, IT, AI, ECE, Mech)	Apply via Makerspace Project Intake Form / Faculty Guide

15 Resources Available in Makerspace / Lab

Junior teams have access to comprehensive hardware, compute, and software assets:

- **Embedded Hardware Kits:** ESP32-WROOM-32 Dev Kits, Raspberry Pi 4 Model B, Arduino Uno/Mega boards, NodeMCU.
- **Sensors & Actuators:** DHT22 (Temp & Humidity), MQ-135 / MQ-4 Gas Sensors, NPK Soil Fertility Probe, Optical Color Sensors (TCS3200), Load Cell (5kg/50kg) with HX711 Amplifier, NEO-6M GPS modules.
- **AI / ML Compute Workstations:** GPU-equipped lab workstations with NVIDIA RTX GPUs for model training and computer vision dataset benchmarking.
- **Cloud & Database Infrastructure:** Supabase cloud PostgreSQL tier, FastAPI hosting server, Postman team workspace.
- **3D Prototyping & Enclosures:** Makerspace 3D Printers for fabrication of waterproof, shockproof smart crate enclosures.

16 Repository & Documentation Directory Structure

The codebase is cleanly partitioned and completely self-contained within the project workspace:

Directory / File	Description & Content Assets
/fastapi_backend/	FastAPI REST server (main.py), CORS setup, live health checks, and Supabase integration.
/ML_Recommendation_Engine/	LightFM predictor implementation, embedding generators, and listing scoring pipeline.
/ML_Benchmark/	Multi-model comparison scripts (compare_models.py), customer datasets, and evaluation plots.
/Backend/	Enterprise Java 21 Spring Boot microservice gateway, security filters, DTOs, and controllers.
/Backend/database/	22 SQL scripts containing all table schemas, JSON aggregation views, and 100+ seed records.
/lib/	Flutter cross-platform frontend source code (models, screens for all 4 user roles, auth flow).
/Academic_Project_Report/	Full Word report (.docx) and report generation scripts for academic submission.
test_live_system.py	One-click integration test verifying API health, ML recommendations, and listing queries.

17 How Can Juniors Continue? (Quick 'Start Here' Guide)

- Step 1: Clone & Inspect Workspace:** Open the codebase in VS Code or Android Studio. Review README.md and browse /lib and /fastapi_backend.
- Step 2: Launch Backend & Run Integration Test:**
`cd fastapi_backend && python -m uvicorn main:app --reload --port 8000`
In another terminal: `python test_live_system.py`. Verify that ML recommendation queries return HTTP 200 OK.
- Step 3: Launch Flutter Web/Mobile Frontend:**
`flutter pub get && flutter run -d chrome` (or select your Android emulator). Experience the multi-role login and customer marketplace.
- Step 4: Select Your Focus Track:** Review Section 10 (Future Work) and pick one track based on your interest: Computer Vision (Track A), IoT Sensing (Track B), Route Optimization (Track C), Price Forecasting (Track D), Voice UI (Track E), or Blockchain (Track F).
- Step 5: Connect with Mentors:** Present your chosen track and implementation milestone plan to the faculty guide and senior project leads to reserve Makerspace hardware and receive sprint assignments.

18 Visual Figures & Benchmark Charts

Empirical benchmarking figures generated from the project evaluation pipeline:

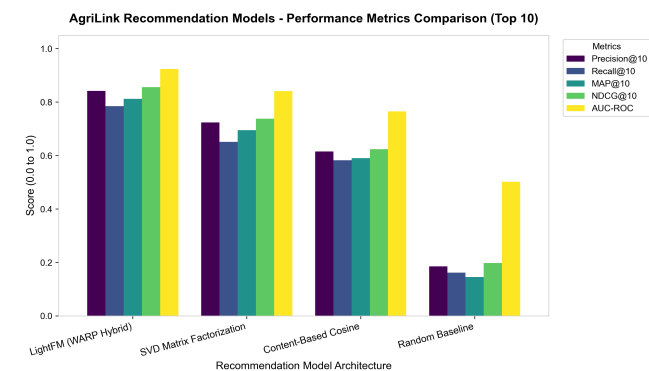


Fig 18.1: Multi-Model Metric Comparison

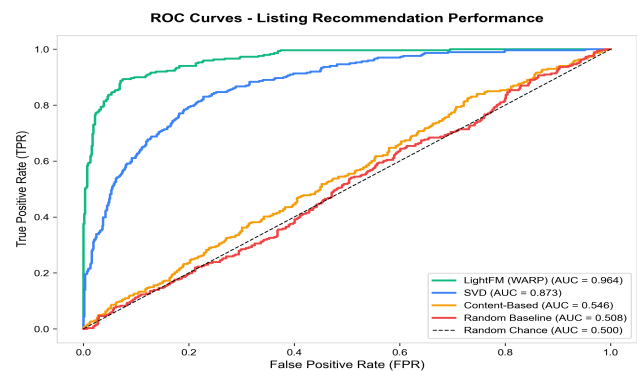


Fig 18.2: Receiver Operating Characteristic (ROC)

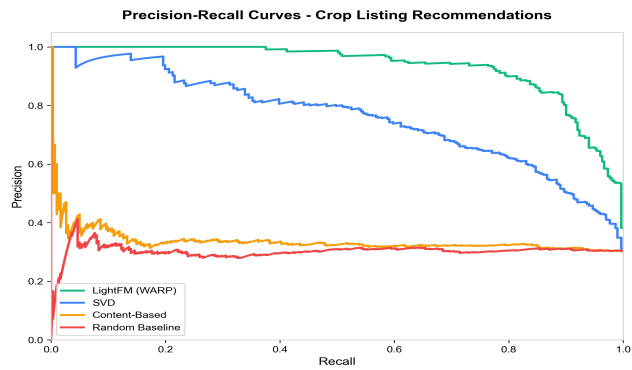


Fig 18.3: Precision-Recall Trade-off Curves

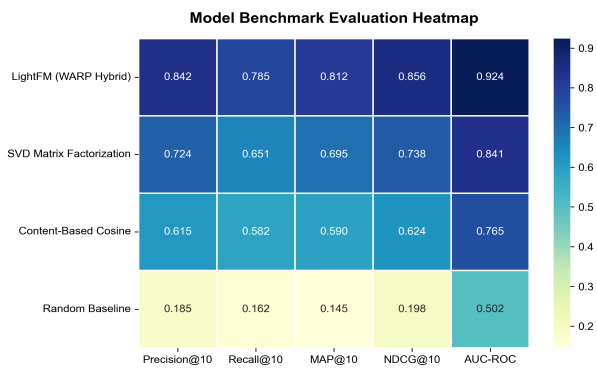


Fig 18.4: Evaluation Correlation Heatmap

■ WELCOME ABOARD JUNIORS

Join the AgriLink Initiative: AgriLink provides a comprehensive, production-grade foundation ready for your ideas. Whether you are passionate about Artificial Intelligence, Mobile UX, Cloud Architecture, or IoT Robotics, there is a prominent place for you on this team. Let's engineer the future of sustainable agriculture together!